Litchi Pack House Process Comparison

Laughing Waters Litchi Farm • Mpumalanga, South Africa

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This table provides a side-by-side comparison of how fruit is handled across the different market channels at the pack house. It is designed to support the development of digital tools by clearly showing where processes diverge and what data needs to be captured for each route.

Stage	Export (USA)	Export (EU)	Local Market	Juice Fruit	Pre-Pack Route
Harvest & Selection	Best quality fruit selected, especially mornings. Block recorded.	Best quality fruit selected. Block recorded for traceability.	Good appearance, reasonable size, minimal marks. Influenced by daily demand & weather.	Smallest fruit (< ~30 mm) and lower grade/irregular fruit.	Premium sizes (esp. Double XL, XL) with best appearance.
Intake & Documentation	Full documentation: block, quantity, time. High traceability required.	Full documentation: block, quantity, time. Traceability important.	Basic documentation of block and quantity. Less strict traceability.	Basic block and quantity recorded. Minimal additional data needed.	Same intake as other routes. Pre-pack decision made at the grading & sizing table.
Grading & Sizing	Strict grading. Premium sizes prioritised for export.	Strict grading. Premium sizes prioritised for EU buyers.	Grading for acceptable appearance. Preference for larger sizes.	Fruit < ~30 mm or heavily marked separated into juice bins at the grading table.	Graded & sized first. Best fruit diverted to the pre-pack table.
Sulphur Fumigation	Yes – 425 g in tent, loaded by pallet (~24 crates) after grading. Critical for long-distance colour retention.	Yes – 425 g in tent, loaded by pallet (~24 crates) after grading. Critical for long journey to Europe.	Generally not required (short supply chain).	No – no treatment applied.	**No** – local market only; no chemical treatment.
Acid Bath (Suurbad)	Yes – one of the two acid baths, after sulphur fumigation, to restore the red colour.	Yes – one of the two acid baths, after sulphur fumigation, to restore the red colour.	Yes – straight from grading to the acid bath (no sulphur).	No – diverted at the grading table.	**No** – diverted at the grading table, no chemical treatment.
Packing Method	Standard export cartons. Built into pallets per market specs.	Standard export cartons. Pallets built for EU requirements.	Simpler cartons or packaging for local wholesalers/markets.	Collected in bins (not packed into fresh market cartons).	Punnets (180–500 g) packed at the dedicated table, then into cartons.
Labelling & Traceability	Full labelling + barcodes. Block number on cartons where possible. High traceability.	Full labelling + barcodes. Block info important. Extra weighing often done for container prep.	Basic labelling with farm details. No strict block traceability required on every carton.	Minimal labelling – mainly bin identification.	Labels on punnets and cartons. Barcodes applied. Farm details as for local market.
Cold Storage	Yes – required until PPECB clearance.	Yes – required until PPECB clearance.	Limited or none required.	Minimal – bins stored until dispatch.	Limited or none (local market).
PPECB Inspection	Yes – mandatory for USA access. Strict checks.	Yes – required for EU export compliance.	No.	No.	**No** – local market only.
Dispatch & Logistics	**Sea:** truck to Johannesburg → container → Cape Town harbour → sea to USA. **Air:** truck to OR Tambo → air freight to USA.	**Sea:** truck to Johannesburg → container → Cape Town harbour → sea to Europe. **Air:** truck to OR Tambo → air freight to Europe.	Direct truck to local markets or Johannesburg wholesalers. Fast and flexible.	Direct truck to juice processors/buyers. Simplest logistics.	Direct truck to premium local markets or Johannesburg wholesalers.
Main Purpose / Value	Highest value export channel. Strictest requirements.	High value export to Europe. Similar requirements to USA with some operational differences.	Good volume outlet. More flexible and responsive to daily market conditions.	Recover value from fruit that cannot be sold fresh. Lowest processing cost.	Value-adding step for premium fruit. Meets local market preference for pre-packed product.

Step order: intake → grading & sizing table (pre-pack and juice fruit diverted here) → sulphur fumigation (export only) → acid bath (export and local) → packing → cold storage & PPECB (export) → dispatch.